

- 1 Sand dunes are important ecosystems that are found in many regions of the world. Sand dune ecosystems can be damaged by natural events, such as extreme weather conditions, and by human activities including farming and tourism.

Fig. 1.1 is a map of part of a national park in a coastal area of northern Europe. The map shows the range of ecosystems present, including two types of sand dune ecosystem: yellow dune and grey dune.

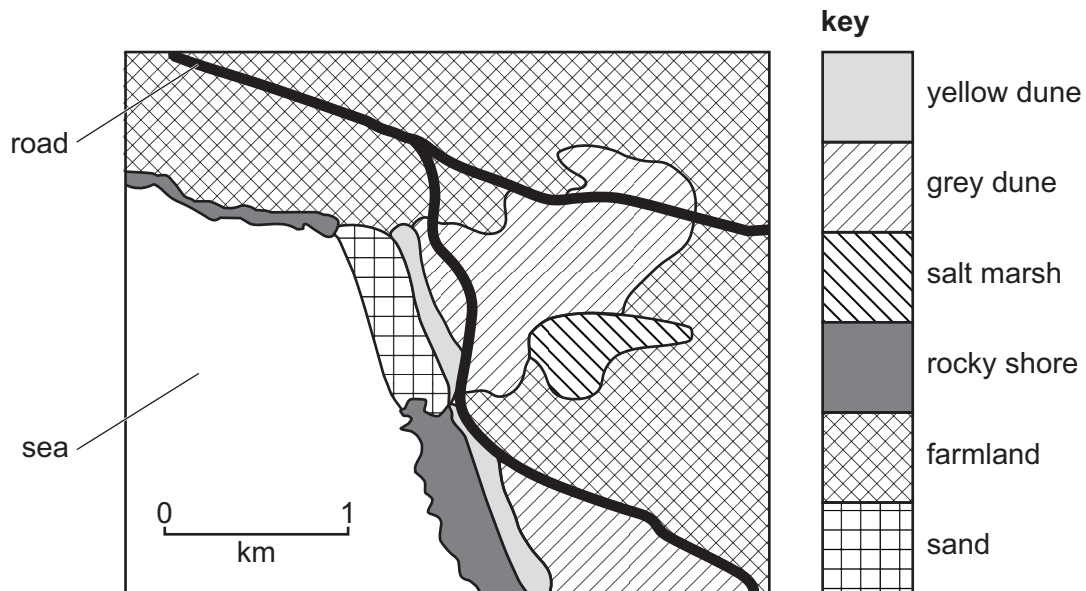


Fig. 1.1

A biologist decided to study the grey dune ecosystem. The biologist visited the ecosystem in the summer, in the month of June.

- (a) The grey dune ecosystem is a habitat for the grey bush cricket, *Platycleis albopunctata*.

Fig. 1.2 shows a grey bush cricket.



Fig. 1.2



The biologist wanted to estimate the population size of the grey bush cricket in the grey dune ecosystem.

The biologist used a sweep net to catch grey bush crickets. The biologist moved the sweep net through the plants growing in the grey dune ecosystem. Any grey bush crickets on the plants were caught in the net.

Fig. 1.3 shows a biologist using a sweep net.



Fig. 1.3

Describe the mark-release-recapture method that the biologist could use to estimate the population size of the grey bush cricket.

.....

.....

.....

.....

.....

.....

..... [3]

- (b) The biologist also estimated the species diversity of the plants in the grey dune ecosystem.

The biologist counted the number of individual plants of each species in a small area and calculated Simpson's index of diversity (D).

The formula for Simpson's index of diversity (D) is:

$$D = 1 - \left(\sum \left(\frac{n}{N} \right)^2 \right)$$

key to symbols:

n = number of individuals of each species present in the sample

N = the total number of all individuals of all species present in the sample

Table 1.1 shows the results.

Complete Table 1.1 to calculate Simpson's index of diversity (D) to **three** significant figures.

Table 1.1

species	number of individuals (<i>n</i>)	$\frac{n}{N}$	$\left(\frac{n}{N}\right)^2$
marram grass	8		
red fescue	38		
creeping buttercup	3		
red clover	2		
heath dog violet	3		
lady's bedstraw	9		
restharrow	20		
ribwort plantain	1		
salad burnet	12		
spiked speedwell	4		
<i>N</i> = 100		$\Sigma\left(\frac{n}{N}\right)^2 = \dots\dots\dots$	
Simpson's index of diversity (<i>D</i>) =			

[3]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

- (c) The national park managers introduced some conservation practices to protect the grey sand dune ecosystem and maintain biodiversity.

The biologist decided to investigate changes to the ecosystem during the ten-year period after the conservation practices were introduced.

- (i) The biologist monitored the light intensity and temperature in the grey dune ecosystem.

Suggest **two other** abiotic factors the biologist could monitor during the ten-year period.

.....
..... [2]

- (ii) Describe a method the biologist could use to investigate changes in the species diversity of the plants in the grey dune ecosystem during the ten-year period after the conservation practices were introduced.

Your method should be set out in a logical order and be detailed enough to allow another person to follow it.

Details of how to calculate Simpson's index of diversity should **not** be included.

.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [7]

[Total: 15]